

Lecture 2: Conditional Probability, Independence

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- 1 Conditional probability::: {.callout-note icon=false}## Definition: Conditional probabilityFor events A, B with $P(B) > 0$,

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

::: . :::: {.callout-warning icon=false}## FactMultiplication rule: $P(A \cap B) = P(B)P(A | B)$, extending to chains of events.:—::: {.callout-warning icon=false}## Theorem: Law of Total ProbabilityIf $B_1, \dots, B_n$ partition Ω , then for any event A ,

$$P(A) = \sum_{i=1}^n P(B_i) P(A | B_i).$$

:::## Independence::: {.callout-note icon=false}## Definition: Independent events A and B are independent if $P(A \cap B) = P(A)P(B)$, equivalently $P(A | B) = P(A)$.:.. . :::: {.callout-warning icon=false}## FactIndependence is *not* the same as disjointness: two events of positive probability can never be both independent and disjoint.:—# The Monty Hall problemThree doors, one hides a car. You pick a door; the host (who knows where the car is) opens a *different* door revealing a goat. Should you switch? 2

1.1 Reading & HomeworkReading: §1.3-1.5 (Bertsekas & Tsitsiklis, *Introduction to Probability*). . 2

—title: “Lecture 2: Conditional Probability, Independence”author: “Jamie Haddock”format: revealjs: output-file: Lecture2_slides slide-number: true preview-links: auto logo: figs/hmc.png css: input/slides.css incremental: true smaller: true code-fold: true embed-resources: true html: code-fold: true embed-resources: true pdf: documentclass: article toc: true number-sections: true geometry: - top=1in - left=1in - bottom=1in - right=1informat-links: falsejupyter: python3filters: - input/remove-pause.luaexecute: echo: true eval: true—

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::: . :::: {.callout-warning icon=false}## FactMultiplication rule: $P(A \cap B) = P(B)P(A | B)$, extending to chains of events.:—::: {.callout-warning icon=false}## Theorem: Law of Total ProbabilityIf $B_1, \dots, B_n$ partition Ω , then for any event A ,

$$P(A) = \sum_{i=1}^n P(B_i) P(A | B_i).$$

Independence: Independent events A and B are independent if $P(A \cap B) = P(A)P(B)$, equivalently $P(A \mid B) = P(A)$.
Fact: Independence is *not* the same as disjointness: two events of positive probability can never be both independent and disjoint.
The Monty Hall problem: Three doors, one hides a car. You pick a door; the host (who knows where the car is) opens a *different* door revealing a goat. Should you switch?

```
def monty_hall_trial(switch):    car_door = rng.integers(0, 3)    choice = rng.integers(0, 3)    remain
```

Exercise Use the law of total probability to compute $P(\text{win} \mid \text{switch})$ by conditioning on which door hides the car.

Answer:

Conditioning on the car's door (each with probability $1/3$): if you initially picked the car (prob $1/3$), switching loses. If you didn't (prob $2/3$), the host's forced reveal means switching always wins. So $P(\text{win} \mid \text{switch}) = \frac{1}{3} \cdot 0 + \frac{2}{3} \cdot 1 = \frac{2}{3}$.

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